

Advanced Data Management

Mappings

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Outline of the course

1. Introduction to propositional logic
 2. Introduction to First-order Logic
 3. Relational Calculus
 4. Information Integration **S**ystems (IIS)
 5. Logical formalization of IISs
 6. **Mapping between Global Schema e Data Sources**
 7. Incomplete information databases
 8. Query answering over IISs
 9. **O**ntology-**B**ased **D**ata **M**anagement (OBDM)
 10. Description Logic Ontologies
 11. Query answering in ontologies
 12. Query answering in OBDM
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Information Integration Systems (recap)

Formalization of Information Integration Systems

- The architecture of an Information Integration Systems (I.I.S.) can be defined using **First-Order Logic**
 - However, we need to make some simplifying assumptions
- Data sources are represented as one single relational database
 1. No syntactic heterogeneity between sources
 2. A single logical theory (**source schema**) describes valid data sources
 3. Source schemas may contain constraints (FOL sentences)
- The result of the integration is a relational database.
 1. No syntactic heterogeneity between sources and the result of integration
 2. A single logical theory (**global schema**) describes the result of the integration
 3. Global schemas may contain constraints (FOL sentences)

Information Integration Specification - Syntax

An information integration specification is a triple $J = \langle G, M, S \rangle$, where:

- **G** is the **global schema**
 - A relational database schema that describes how data resulting from the integration process should look like
 - A database **B** for **G** will be called **global database**, i.e., we assume that $B \models G$
- **S** is the **source schema**
 - A relational database schema describing the data sources such that $G \cap S = \emptyset$
 - A database **D** for **S** will be called **source database**, i.e., we assume that $D \models S$
- **M** is the **mapping** between **source schema S** and **global schema G**
 - A set of mapping assertions $m = \langle q_S, q_G \rangle$, where q_S is a FOL query over **S** and q_G is a FOL query over **G**. q_S and q_G have the same arity!

Information Integration System - Semantics

An information integration system is a pair $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is an information integration specification and $\mathbf{D}$ is a **source database** for $\mathbf{S}$

A global database $\mathbf{B}$ **satisfies** the mapping $\mathbf{M}$ with respect to $\mathbf{D}$, denoted by $(\mathbf{B}, \mathbf{D}) \models \mathbf{M}$, if $q_S(\mathbf{D}) \subseteq q_G(\mathbf{B})$ for each $m \in \mathbf{M}$ (**sound source assumption**)

The semantics of the information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$ is the set of **global databases** that satisfy the **mapping M** w.r.t. to the **source database D**.

Formally:

$$\text{sem}(\mathbf{J}, \mathbf{D}) = \{ \mathbf{B} \mid \mathbf{B} \text{ is a global database s.t. } (\mathbf{B}, \mathbf{D}) \models \mathbf{M} \}$$

This semantics is based on **sound mappings**, but different semantics for mappings are possible.

Querying an Information Integration System

Consider an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ and $\mathbf{D}$ is a **source database** for $\mathbf{S}$, and query q over $\mathbf{G}$

The **certain answers** for q w.r.t. $\langle \mathbf{J}, \mathbf{D} \rangle$ are those tuples of constants that occur in the answers for q over **every possible global database** in the semantics of $\langle \mathbf{J}, \mathbf{D} \rangle$.

Formally:

$$\text{cert}(q, \mathbf{J}, \mathbf{D}) = \{(c_1, \dots, c_n) \mid (c_1, \dots, c_n) \in q(\mathbf{B}) \text{ for each } \mathbf{B} \in \text{sem}(\mathbf{J}, \mathbf{D})\}$$

That is, the *intersection* of the answers for q over all the **global database** in $\text{sem}(\mathbf{J}, \mathbf{D})$

Different Forms of Mappings

We introduced the so called **sound mapping** semantics

- Why do we like it?
- Are there any other possible semantics?

What kind of queries can be used to specify mappings?

- Is there any limitation?
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Semantics for Mappings

Sound Mappings

The **sound-mapping semantics**: Information at the **sources** may be partial but it is never wrong (i.e., it is valid information for the **global database**)

EX: The sources contain tables **Customers** and **Purchases**. The names of customers there represent actual **people** BUT there may be more **people** than those in the **sources**.

Sound Mappings and Logic

Assume an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, s.t. $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is and $\mathbf{D}$ is a **source database** for $\mathbf{S}$

The sound-mapping semantics of $\langle \mathbf{J}, \mathbf{D} \rangle$ are defined as follows:

$$\text{sem}(\mathbf{J}, \mathbf{D}) = \{ \mathbf{B} \mid q_{\mathbf{S}}(\mathbf{D}) \subseteq q_{\mathbf{G}}(\mathbf{B}) \text{ for each } \mathbf{m} \in \mathbf{M} \}$$

A mapping assertions $\mathbf{m}$ is of the form $\langle q_{\mathbf{S}}, q_{\mathbf{G}} \rangle$, where $q_{\mathbf{S}} = \{ \bar{x} \mid \varphi_{\mathbf{S}}(\bar{x}) \}$ is a FOL query over $\mathbf{S}$ and $q_{\mathbf{G}} = \{ \bar{x} \mid \varphi_{\mathbf{G}}(\bar{x}) \}$ is a FOL query over $\mathbf{G}$.

Such mappings have a natural logical formulation:

$$\forall \bar{x}. \varphi_{\mathbf{S}}(\bar{x}) \rightarrow \varphi_{\mathbf{G}}(\bar{x})$$

Sound Mappings: Example – the source DB

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Sound Mappings: Example – the Global Schema

Person

Id	Name
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Sound Mappings: Example – the IIS

Assume $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ with:

- $\mathbf{G} = \{Person_{/2}\}$
- $\mathbf{S} = \{Customers_{/3}\}$
- $\mathbf{M} = \{m\}$ where:
 - $m = \langle \{(n) \mid \exists d. \exists r. Customers(n, d, r)\}, \{(n) \mid \exists i. Person(i, n)\} \rangle$

and its logical formalization ρ defined as follows:

- $\rho : \forall n. \exists d. \exists r. Customers(n, d, r) \rightarrow \exists i. Person(i, n)$

Assume a **source database** $\mathbf{D}$ as defined previously.

We have that a **global database** $\mathbf{B}$ satisfies $\mathbf{M}$ w.r.t. $\mathbf{D}$, written $(\mathbf{B}, \mathbf{D}) \models \mathbf{M}$, if $q_S(\mathbf{D}) \subseteq q_G(\mathbf{B})$.

We can prove that $q_S(\mathbf{D}) \subseteq q_G(\mathbf{B})$ if and only if $(\mathbf{B}, \mathbf{D}) \models \rho$

Sound Mappings and Logic: Example

$\forall n. \exists d. \exists r. \text{Customers}(n, d, r) \rightarrow \exists i. \text{Person}(i, n)$

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Person

Id	Name
1	Amber
2	Rose

OK

Sound Mappings and Logic: Example

$\forall n. \exists d. \exists r. \text{Customers}(n, d, r) \rightarrow \exists i. \text{Person}(i, n)$

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Person

Id	Name
1	Amber
2	Rose
3	Kelly

OK

Sound Mappings and Logic: Example

$\forall n. \exists d. \exists r. \text{Customers}(n, d, r) \rightarrow \exists i. \text{Person}(i, n)$

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Person

Id	Name
Something	Amber
Something	Rose
Something	Something

OK

Sound Mappings and Logic: Example

$\forall n. \exists d. \exists r. \text{Customers}(n, d, r) \rightarrow \exists i. \text{Person}(i, n)$

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Person

Id	Name
1	Amber
2	Bob

NO: Rose is missing!

Fully Logical Formalization of Information Integration Specifications - Syntax

An information integration specification is a triple $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$, where:

- $\mathbf{G}$ is the **global schema**: a logical theory over a relational alphabet A_G
- $\mathbf{S}$ is the **source schema**: a logical theory over a relational alphabet A_S
- $\mathbf{M}$ is the **mapping** between $\mathbf{S}$ and $\mathbf{G}$: a logical theory over the relational alphabet $A_S \cup A_G$.

Fully Logical Formalization of Information Integration Specifications - Semantics

Assume an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is an information integration specification and $\mathbf{D}$ is a **source database** for $\mathbf{S}$

The semantics of $\langle \mathbf{J}, \mathbf{D} \rangle$ is:

$$\text{sem}(\mathbf{J}, \mathbf{D}) = \{ \mathbf{B} \mid \mathbf{B} \text{ is a global database s.t. } (\mathbf{B}, \mathbf{D}) \models \mathbf{M} \}$$

Where $(\mathbf{B}, \mathbf{D}) \models \mathbf{M}$ means that $(\mathbf{B}, \mathbf{D}) \models \rho_m$ for each $m \in \mathbf{M}$, where ρ_m is the logical formalization of m .

Observe that in this formalization we are not forced to use sound mappings!

- What kind of mappings are reasonable?

Fully Logical Formalization of Information Integration Specifications - Queries

Consider an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ is an information integration specification and $\mathbf{D}$ is a **source database** for $\mathbf{S}$, and a query q over $\mathbf{G}$

The **certain answers** for q w.r.t. $\langle \mathbf{J}, \mathbf{D} \rangle$ consist in the set:

$$\text{cert}(q, \mathbf{J}, \mathbf{D}) = \{(c_1, \dots, c_n) \mid (c_1, \dots, c_n) \in q(\mathbf{B}) \text{ for each } \mathbf{B} \in \text{sem}(\mathbf{J}, \mathbf{D})\}$$

Since $\text{sem}(\mathbf{J}, \mathbf{D})$ is well defined also in the case of a fully logical formalization, we can use the **standard notion of certain answers**.

Other Possible Semantics for Mappings

The sound-mapping semantics. Information at the sources may be partial but it is never wrong.

What are the fundamental assumptions?

- Possibly partial information
- Always correct information

Other options for $\langle q_S, q_G \rangle$?

- Information may be *incorrect*, i.e., q_S describes more than the answers for q_G
- Information is exact, i.e., q_S describes all the possible answers for q_G

Sound, Complete, and Exact Mappings

Let $\langle \mathbf{J}, \mathbf{D} \rangle$ be an information integration system, where $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ and $\mathbf{D}$ is a **source database** for $\mathbf{S}$

The sound-mapping semantics: **sources** contain partial but never wrong information regarding the **global schema**

$$\text{sem}(\mathbf{J}, \mathbf{D}) = \{ \mathbf{B} \mid q_S(\mathbf{D}) \subseteq q_G(\mathbf{B}) \text{ for each mapping } \langle q_S, q_G \rangle \in \mathbf{M} \}$$

The complete-mapping semantics: **sources** contain complete but possibly wrong information

$$\text{sem}(\mathbf{J}, \mathbf{D}) = \{ \mathbf{B} \mid q_S(\mathbf{D}) \supseteq q_G(\mathbf{B}) \text{ for each mapping } \langle q_S, q_G \rangle \in \mathbf{M} \}$$

The exact-mapping semantics: **sources** contain exact information regarding the **global schema**

$$\text{sem}(\mathbf{J}, \mathbf{D}) = \{ \mathbf{B} \mid q_S(\mathbf{D}) \equiv q_G(\mathbf{B}) \text{ for each mapping } \langle q_S, q_G \rangle \in \mathbf{M} \}$$

Other Semantics for Mappings: Example

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Other Semantics for Mappings: Example

Assume $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ with:

- $\mathbf{G} = \{Person_{/2}\}$
- $\mathbf{S} = \{Customers_{/3}\}$
- $\mathbf{M} = \{m = \langle q_S, q_G \rangle\}$ where:
 - $q_S = \{ (n) \mid \exists d. \exists r. Customers(n, d, r) \}$
 - $q_G = \{ (n) \mid \exists i. Person(i, n) \}$

We examine the different semantics of for m

Other Semantics for Mappings: Example

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Person

Id	Name
1	Amber
2	Rose

This is a model under all semantics

Other Semantics for Mappings: Example

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Person

Id	Name
1	Amber
2	Rose
3	Kelly

This is a model only under the sound semantics

Other Semantics for Mappings: Example

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Person

Id	Name
1	Amber
2	Rose
3	Rose

This is a model under all semantics

Other Semantics for Mappings: Example

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam
Rose	10/20/1930	Constantinople

Person

Id	Name
1	Amber

This is a model only under complete semantics

Logical Formalization of the Different Semantics

Assume a mapping assertion $m = \langle q_S, q_G \rangle$ where q_S and q_G are expressed as FOL queries:

$$q_S = \{ \bar{x} \mid \varphi_S(\bar{x}) \} \text{ and } q_G = \{ \bar{x} \mid \varphi_G(\bar{x}) \};$$

Sound Semantics of m

$$- \forall \bar{x}. \varphi_S(\bar{x}) \rightarrow \varphi_G(\bar{x})$$

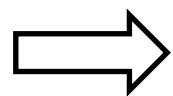
Complete Semantics of m

$$- \forall \bar{x}. \varphi_S(\bar{x}) \leftarrow \varphi_G(\bar{x})$$

Exact Semantics of m

$$- \forall \bar{x}. \varphi_S(\bar{x}) \rightarrow \varphi_G(\bar{x})$$

$$- \forall \bar{x}. \varphi_S(\bar{x}) \leftarrow \varphi_G(\bar{x})$$



$$\forall \bar{x}. \varphi_S(\bar{x}) \leftrightarrow \varphi_G(\bar{x})$$

Exact Mappings: Some Counterintuitive Behavior

- One could be tempted to think that **exact** mappings are the most reasonable semantics for Information Integration Systems.
- Actually, **complete** and **exact** mappings can have a very **counterintuitive behavior**
 - Information at the **sources** is influenced by information in the target schema!
 - Simple assertions may give rise to **inconsistent systems (i.e., systems with no semantics)**

Exact Mappings: Example

Let $\mathbf{J} = \langle \mathbf{G}, \mathbf{M}, \mathbf{S} \rangle$ be such that

- $\mathbf{G} = \{Person_{/2}\}$
- $\mathbf{S} = \{Customers_{/3}, Purchases_{/3}\}$
- $\mathbf{M} = \{m_1, m_2\}$ where:
 - $m_1 : \forall n. \exists d. \exists r. Customers(n, d, r) \leftrightarrow \exists i. Person(i, n)$
 - $m_2 : \forall n. \exists i. \exists p. Purchases(i, n, p) \leftrightarrow \exists i. Person(i, n)$

Let $\mathbf{D}$ be a **source database** as follows:

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam

Purchases

ID	Customer Name	Product
O002	Dylan	P001

We can show that that $\mathbf{sem}(\mathbf{J}, \mathbf{D}) = \emptyset$

Exact Mappings: Example

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam

Person

Id	Name
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Purchases

ID	Customer Name	Product
O002	Dylan	P001

$m_1 : \forall n. \exists d. \exists r. \text{Customers}(n, d, r) \leftrightarrow \exists i. \text{Person}(i, n)$

$m_2 : \forall n. \exists i. \exists p. \text{Purchases}(i, n, p) \leftrightarrow \exists i. \text{Person}(i, n)$

Exact Mappings: Example

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam

Person

Id	Name
1	Amber

Purchases

ID	Customer Name	Product
O002	Dylan	P001

$m_1 : \forall n. \exists d. \exists r. \text{Customers}(n, d, r) \leftrightarrow \exists i. \text{Person}(i, n)$

$m_2 : \forall n. \exists i. \exists p. \text{Purchases}(i, n, p) \leftrightarrow \exists i. \text{Person}(i, n)$

Violate m_2

Exact Mappings: Example

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam

Person

Id	Name
2	Dylan

Purchases

ID	Customer Name	Product
O002	Dylan	P001

$m_1 : \forall n. \exists d. \exists r. \text{Customers}(n, d, r) \leftrightarrow \exists i. \text{Person}(i, n)$

$m_2 : \forall n. \exists i. \exists p. \text{Purchases}(i, n, p) \leftrightarrow \exists i. \text{Person}(i, n)$

Violate m_1

Exact Mappings: Example

Customers

Name	DOB	Residency
Amber	10/20/2030	New Amsterdam

Person

Id	Name
1	Amber
2	Dylan

Purchases

ID	Customer Name	Product
O002	Dylan	P001

$m_1 : \forall n. \exists d. \exists r. \text{Customers}(n, d, r) \leftrightarrow \exists i. \text{Person}(i, n)$

$m_2 : \forall n. \exists i. \exists p. \text{Purchases}(i, n, p) \leftrightarrow \exists i. \text{Person}(i, n)$

Violate both m_1 and m_2

Query Answering Under Sound Mappings

From now on we focus only on **sound mappings** and assume their logical formalization.

We have a problem:

Thm: Given a database **D**, an information integration system $\langle \mathbf{J}, \mathbf{D} \rangle$, and a FOL query **q**, computing $\text{cert}(\mathbf{q}, \mathbf{J}, \mathbf{D})$ is **undecidable** (actually to be undecidable is the *associated decision problem*)

The theorem above tells us that we cannot expect to answer queries over and Information integration system with **arbitrarily expressive mappings**.

We need to impose further restrictions on the languages involved

Complexity of Query Answering

Thm: Given a database D , an information integration system $\langle J, D \rangle$, and a FOL query q , computing $\text{cert}(q, J, D)$ is **undecidable** (actually to be undecidable is the *associated decision problem* – see later).

Proof: The undecidability follows from the undecidability of the validity problem in FOL. Assume to have:

- a mapping $M = \{\forall x. T(x) \rightarrow \psi(x)\}$ with $\psi = (\varphi \rightarrow R(b))$.
- a query $q = \exists x. R(x)$
- a source database D such that $D(T) = \{a\}$

We have that:

- If φ is a tautology then $\langle b \rangle \in B(R)$ for each $B \in \text{sem}(J, D)$.
 - We conclude $\text{cert}(q, J, D) \neq \emptyset$
- If φ is not a tautology then $B(R) = \emptyset$ for some $B \in \text{sem}(J, D)$.
 - We conclude $\text{cert}(q, J, D) = \emptyset$

Restricting the Language

We need to impose further restrictions on the languages involved.

The restriction that we will adopt (see the slides on query answering) is to assume mappings that only use **conjunctive queries**

$$\forall \bar{x} \quad \varphi(\bar{x}) \rightarrow \psi(\bar{x})$$

where both φ and ψ are **existential conjunctions**, i.e., formulae of the form

$$\exists y_1, \dots, \exists y_n \, P_1(\bar{z}_1) \wedge \dots \wedge P_k(\bar{z}_k)$$

Different Forms of Mappings: GAV, LAV, and GLAV

Different Forms of Mappings

Inside **sound conjunctive mappings**, we distinguish three important and widely used classes:

The **global schema** is defined in terms of the data at the **sources**

- **GAV: Global-As-View**

Data at sources is defined in terms of the **global schema**

- **LAV: Local-As-View**

A mixed and more powerful approach

- **GLAV: Global-Local-As-View**
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Global-As-View

A GAV mapping describes the **global schema G** in terms of the **source schema S**

A GAV **mapping assertion** is a pair $\langle q_S, g \rangle$ where q_S is a FOL query over the **source schema** and $g \in G$ is a **single atom** of the **global schema**.

Logical formalization of GAV **mapping assertions**

$$\forall \bar{x} . \varphi_S(\bar{x}) \rightarrow g(\bar{x})$$

A **GAV mapping M** is a finite set of GAV mapping assertions

Global As View – Observations

- GAV mappings provide direct information on what data populates the relations of the **global schema**.
- Elements of the **global schema** are considered **views** over the data sources (**Global-As-View**).
- No direct method to cope with incompleteness at the sources.
 - No quantification in the right-hand side of mapping assertions
- GAV is a very natural and simple way to define a mapping.
 - Pretty much the same as defining views over a relational database.

Example of GAV Mapping

- **Global Schema G** containing:
 - **Movie(Title, Year, Director)**
 - **European(Director)**
 - **Review(Title, Critique)**
- **Source Schema S** containing:
 - **r1(Title, Year, Director)**: Movies since 1960 with European directors
 - **r2(Title, Critique)**: Reviews of movies since 1990

GAV Mapping $M=\{m_1, m_2, m_3\}$:

- $m_1: \forall t, y, d. \mathbf{r_1(t, y, d)} \rightarrow \mathbf{Movie(t, y, d)}$
- $m_2: \forall d. \exists y, t. \mathbf{r_1(t, y, d)} \rightarrow \mathbf{European(d)}$
- $m_3: \forall t. \exists t, r. \mathbf{r_2(t, r)} \rightarrow \mathbf{Review(t, r)}$

Local As View

LAV mappings describe the **source schema** in terms of the **global schema**

A LAV **mapping assertion** is a pair $\langle s, q_G \rangle$ where q_G is a **FOL query** over the **global schema** and $s \in \mathbf{S}$ is a **single atom** of the **source schema**.

Logical formalization of LAV mapping assertions

$$\forall \bar{x} \ s(\bar{x}) \rightarrow \varphi_G(\bar{x})$$

A LAV **mapping** is a **finite** set of LAV mapping assertions.

- We assume that every element of the source schema is **defined exactly once**

Local As View – Observations

- LAV mappings **do not provide** direct information on what data populates the **global schema**.
 - Single elements of the **global schema** may not be explicitly defined
- **Mappings** are defined considering **data sources** as **views** over the **global schema** (**Local-As-View**).
- They provide a direct method to cope with **incompleteness** at the **sources**.
 - Quantification can appear in the right-hand side of mapping assertions
- LAV is a sophisticated way to define a mapping: modelling requires attention

Example of LAV Mapping

- **Global Schema G** containing:
 - **Movie(Title, Year, Director)**
 - **European(Director)**
 - **Review(Title, Critique)**
- **Source Schema S** containing:
 - **r1(Title, Year, Director)**: Movies since 1960 with European directors
 - **r2(Title, Critique)**: Reviews of movies since 1990

LAV Mapping M={m₁,m₂}:

- **m₁**: $\forall t, y, d. \mathbf{r_1(t, y, d)} \rightarrow \mathbf{Movie(t, y, d)} \wedge \mathbf{European(d)}$
- **m₂**: $\forall t, r. \mathbf{r_2(t, r)} \rightarrow \exists y, d. \mathbf{Review(t, r)} \wedge \mathbf{Movie(t, y, d)}$

GAV vs LAV Mapping

- **GAV**

- Quality depends on how well we can compile the **sources** into the **global schema**
- Changing on the **sources** may reflect heavily on the mapping
- Query processing is very easy and based on some sort of *unfolding* (see later)

- **LAV**

- Quality depends on how well we can characterize the **sources** using the **global schema**
 - Highly modularity and extensibility. In case of changes at the **sources** we can simply redefine the portion that has changed.
 - Query processing is not straightforward (see later)
-

Global and Local As View

- GLAV can define the **source schema** using the **global schema** and vice versa (**Global-And-Local-As-View**) and define complex constructions
- A **GLAV mapping assertion** is a pair $\langle q_s, q_G \rangle$ where q_G is a FOL query over the **global schema** and q_s is a FOL query over the **source schema**
- Logical formalization GLAV mapping assertions

$$\forall \bar{x} \varphi_s(\bar{x}) \rightarrow \varphi_G(\bar{x})$$

- A GLAV **mapping** is a **finite** set of GLAV mapping assertions.

Global and Local As View – Observations

- GLAV mappings **do not provide** direct information on what data populates the **global schema**.
 - Single elements of the **global schema** may not be explicitly defined
- Direct method to cope with incompleteness at the **sources**.
 - Quantification can appear in both sides of the assertions
- GLAV is a sophisticated way to define a mapping.
 - Modelling requires attention.

Example of GLAV Mapping

- **Global Schema G** containing:
 - Work(Person, Project)
 - Area(Project, Field)
- **Source Schema S** containing:
 - HasJob(Person, Field)
 - Teaches(Professor, Course)
 - In(Course, Field)
 - Get(Researcher, Grant)
 - For(Grant, Project)

GLAV Mapping $M=\{m_1, m_2, m_3\}$:

- $m_1 = \forall r, f. \text{HasJob}(r, f) \rightarrow \exists p. \text{Work}(r, p) \wedge \text{Area}(p, f)$
- $m_2 = \forall r, f. \exists c. \text{Teaches}(r, c) \wedge \text{In}(c, f) \rightarrow \exists p. \text{Work}(r, p) \wedge \text{Area}(p, f)$
- $m_3 = \forall r, p. \exists c. \text{Get}(r, g) \wedge \text{For}(g, p) \rightarrow \text{Work}(r, p)$

Some Technical Consideration

We can think of GLAV mappings as the composition of a LAV and a GAV mapping. Indeed, each GLAV mapping $\langle q_s, q_g \rangle$ can be rewritten as

$$\langle q_s, p \rangle \cup \langle p, q_g \rangle$$

where p is a fresh predicate of the same arity as the two queries q_s and q_g

Example:

$$m_2 = \langle \{(r,f) \mid \exists c. \text{Teaches}(r,c) \wedge \text{In}(c,f)\}, \{(r,f) \mid \exists p. \text{Work}(r,p) \wedge \text{Area}(p,f)\} \rangle$$

- $\langle \{(r,f) \mid \exists c. \text{Teaches}(r,c) \wedge \text{In}(c,f)\}, \{(r,f) \mid p(r,f)\} \rangle$
- $\langle \{(r,f) \mid p(r,f)\}, \{(r,f) \mid \exists p. \text{Work}(r,p) \wedge \text{Area}(p,f)\} \rangle$

From an algorithmic perspective, we can consider LAV and GLAV the **same**.

Exercise

Exercise – Scenario

Description of the Scenario

You are asked to design an information integration system for a company.

Data sources consist of three tables that do **not** contain NULL values:

- **VTables** (id, plateNumber, type, ownerID, ownerType)
This table stores information about vehicles.
 - The attributes **plateNumber** and **ownerID** can contain the value '000' meaning that the information is missing
 - The attribute **type** can be 'motorcycle', 'car', or 'truck'
 - The attribute **ownertype** can be either 'person' or 'company'
- **PTable**(code, age, name, address, phone)
This table stores information about persons
- **CTable**(code, name, address, phone)
This table stores information about companies
- **Constraint:** **VTable(ownerID)** is either a **Person(code)** or a **Company(code)** or 000.
 - $\forall i, p, t_v, o, t_o. VTable(i, p, t_v, o, t_o) \rightarrow \exists x, y, z, w. PTable(o, x, y, z, w) \vee CTable(o, x, y, z) \vee (o = 000)$

Description of the Scenario

You are asked to design an information integration system for a company.

- **Global Schema alphabet** is defined as follows:
 - **Vehicles**_{/1}: set of vehicles.
 - **Motorcycle**_{/1}: set of motorcycles
 - **Car**_{/1}: set of cars.
 - **Owns**_{/2}: Owns(x, y) means x is the owner of y.
 - **hasPlate**_{/2}: hasPlate(x, y) means that vehicle x has plate number y

Requirements.

1. Define suitable axioms for the [global schema](#)
2. Define a suitable GAV mapping.
3. Define a suitable GLAV mapping.
4. Extend the global schema to accommodate the information in the sources.

Solution

Solution: Axioms

1. Define suitable axioms for the global schema

- $\forall x. \text{Motorcycle}(x) \rightarrow \text{Vehicle}(x)$: Motorcycles are Vehicles
- $\forall x. \text{Car}(x) \rightarrow \text{Vehicle}(x)$: Cars are Vehicles
- $\forall x. \text{Motorcycle}(x) \rightarrow \neg \text{Car}(x)$: Motorcycles are not Cars and vice versa
- $\forall x. \text{Vehicle}(x) \rightarrow \exists y. \text{hasPlate}(x, y)$: Every Vehicle has a license plate
- $\forall x. \exists y. \text{hasPlate}(x, y) \rightarrow \text{Vehicle}(x)$: What has a license plate is a vehicle
- $\forall x. \text{Vehicle}(x) \rightarrow \exists y. \text{Owns}(y, x)$: Every Vehicle has a owner
- $\forall x. \forall y. \forall y'. \text{hasPlate}(x, y) \wedge \text{hasPlate}(x, y') \rightarrow (y = y')$: the predicate hasPlate is functional

Solution: GAV Mapping

2. Define a suitable GAV mapping.

- $\forall i, p, t_v, o, t_o. VTable(i, p, t_v, o, t_o) \rightarrow Vehicle(i)$
- $\forall i, p, o, t_o. VTable(i, p, Motorcycle, o, t_o) \rightarrow Motorcycle(i)$
- $\forall i, p, o, t_o. VTable(i, p, Car, o, t_o) \rightarrow Car(i)$
- $\forall i, p, t_v, o, t_o. VTable(i, p, t_v, o, t_o) \wedge (o \neq 000) \rightarrow Owns(o, i)$
- $\forall i, p, t_v, o, t_o. VTable(i, p, t_v, o, t_o) \wedge (p \neq 000) \rightarrow hasPlate(o, i)$

GLAV Mapping

3. Define a suitable GAV mapping

- $\forall i, p, t_v, o, t_o. VTable(i, p, t_v, o, t_o) \rightarrow Vehicle(i)$
- $\forall i, p, o, t_o. VTable(i, p, Motorcycle, o, t_o) \rightarrow Motorcycle(i)$
- $\forall i, p, o, t_o. VTable(i, p, Car, o, t_o) \rightarrow Car(i)$
- $\forall i, p, t_v, o, t_o. VTable(i, p, t_v, o, t_o) \wedge (o = 000) \rightarrow \exists x. Owns(x, i)$
- $\forall i, p, t_v, o, t_o. VTable(i, p, t_v, o, t_o) \wedge (p = 000) \rightarrow \exists x. hasPlate(i, x)$
- $\forall i, p, t_v, o, t_o. VTable(i, p, t_v, o, t_o) \wedge (o \neq 000) \rightarrow Owns(o, i)$
- $\forall i, p, t_v, o, t_o. VTable(i, p, t_v, o, t_o) \wedge (p \neq 000) \rightarrow hasPlate(o, i)$

Solution: Global Schema

4. Extend the global schema to accommodate all the information in the sources.
 - $Vehicles_{/1}$, $Motorcycle_{/1}$, $Car_{/1}$: Set of cars.
 - $Owns_{/2}$, $hasPlate_{/2}$
 - $Person_{/1}$: set of persons
 - $Company_{/1}$: set of companies
 - $hasCode_{/2}$: $hasCode(x, y)$ x has code y
 - $hasAddress_{/2}$: $hasAddress(x, y)$ x has address y
 - $hasPhNum_{/2}$: $hasPhNum(x, y)$ x has phone number y

Solution: Axioms

4. Extend the global schema to accommodate all the information in the sources.

- $\forall x. \text{Motorcycle}(x) \rightarrow \text{Vehicle}(x)$
- $\forall x. \text{Motorcycle}(x) \rightarrow \text{Car}(x)$
- $\forall x. \text{Vehicle}(x) \rightarrow \exists y. \text{hasPlate}(x, y)$
- $\forall x. \exists y. \text{hasPlate}(x, y) \rightarrow \text{Vehicle}(x)$
- $\forall x. \text{Vehicle}(x) \rightarrow \exists y. \text{Owns}(y, x)$
- $\forall x. \forall y. \forall y'. \text{hasPlate}(x, y) \wedge \text{hasPlate}(x, y') \rightarrow (y = y')$

- $\forall x. \text{Person}(x) \rightarrow \exists y. \text{hasCode}(x, y)$
- $\forall x. \text{Company}(x) \rightarrow \exists y. \text{hasCode}(x, y)$
- ...

Credits

The presentations of this module of the course are based on the original slides of Prof. Giuseppe De Giacomo, Prof. Diego Calvanese, Prof. Marco Console, Prof. Maurizio Lenzerini, and Prof. Domenico Lembo.
